Week 5 - Representation and Dimensionality Report

April 2026

1 Executive Summary

This report documents the Week 5 deliverables for representation and dimensionality reduction. The objective is to move from cleaned MovieLens tables to meaningful feature matrices, then validate structure and redundancy through PCA/SVD and supporting visualizations.

- **Project title**: Personalized Movie Discovery and Recommendation Engine (MovieLens 25M)
- **Course**: Semester Project: Domain Discovery, Recommendation, and Graph Intelligence
- **Milestone objective**: Build reproducible feature representations and dimensionality reduction artifacts
- **Report date**: April 2026
- **Status**: Feature matrices, dimensionality reduction, comparison table, and visualizations completed

2 Feature Matrices

Two feature matrices are built at the movie level.

2.1 Numeric and Categorical Matrix

Description

This matrix combines rating aggregates, time-span coverage, and one-hot genres. It captures overall popularity, rating distribution, and catalog category signals in a compact numeric form.

Core columns

- Ratings: `rating_count`, `rating_mean`, `rating_std`, `rating_min`, `rating_max`, `rating_median`
- Temporal: `rating_first_ts`, `rating_last_ts`, `rating_span_days`

- Categorical: one-hot genre columns derived from `movie_genres`
- Tag activity: `tag_count` (count of distinct tags per movie)

Output

- `data/processed/week05/movie_numeric_features.parquet`

2.2 Text Matrix (TF-IDF)

Description

This matrix represents the semantic profile of each movie using aggregated tags and titles. We concatenate `title` and the unique tag list, then apply TF-IDF with English stop-word removal.

TF-IDF configuration

- `max_features = 5000`
- `min_df = 2`
- `stop_words = "english"`

Outputs

- `data/processed/week05/movie_text_tfidf.npz`
- `data/processed/week05/movie_text_tfidf_vocab.json`

2.3 Data Sources

- `data/processed/week03_v1/movies_catalog.parquet`
- `data/processed/week03_v1/movie_genres.parquet`
- `data/processed/week03_v1/ratings_clean.parquet`
- `data/processed/week03_v1/tags_clean.parquet`

3 Dimensionality Reduction

3.1 PCA (Numeric Matrix)

- Input: standardized numeric matrix
- Target variance: 90% cumulative variance
- Output: `data/processed/week05/movie_pca_embeddings.parquet`
- Artifact: `artifacts/week05/pca_variance.csv`

3.2 Truncated SVD (Text TF-IDF)

- Input: sparse TF-IDF matrix
- Components: 200
- Output: `data/processed/week05/movie_svd_embeddings.parquet`
- Artifact: `artifacts/week05/svd_variance.csv`

3.3 t-SNE (Visualization)

- Input: PCA embeddings
- Output: `artifacts/week05/pca_tsne.parquet`

4 Comparison Table

Method	Components	Cumulative Variance	Reconstruction Error
PCA	20	0.9012	0.0988
TruncatedSVD	200	0.3572	0.6373

Table 1: Dimensionality reduction comparison summary from `artifacts/week05/comparison_table.csv`.

5 Visualization Set

The Plotly figures are exported to HTML and PNG.

- `artifacts/week05/pca_cumulative_variance.html`
- `artifacts/week05/svd_cumulative_variance.html`
- `artifacts/week05/tsne_scatter.html`

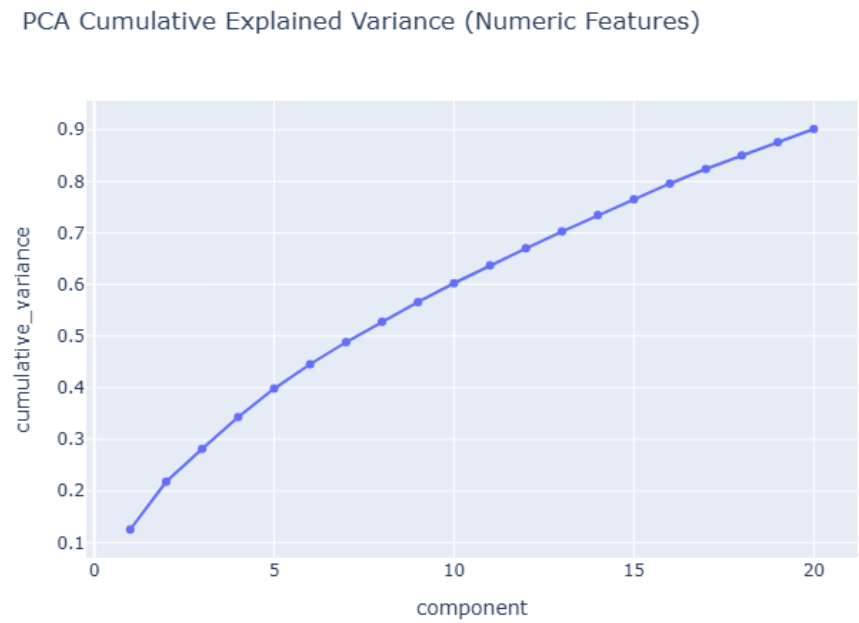

Figure 1: PCA cumulative explained variance.

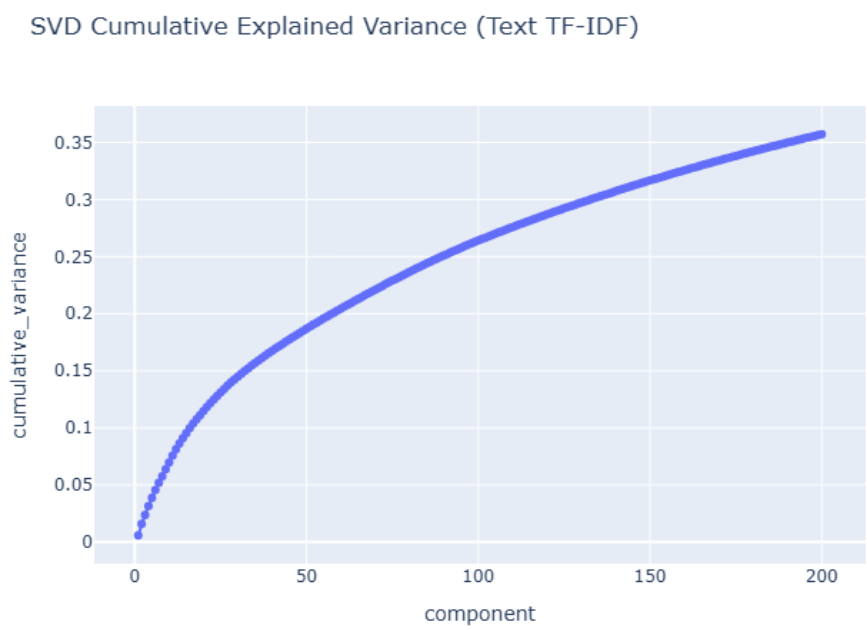

Figure 2: SVD cumulative explained variance.

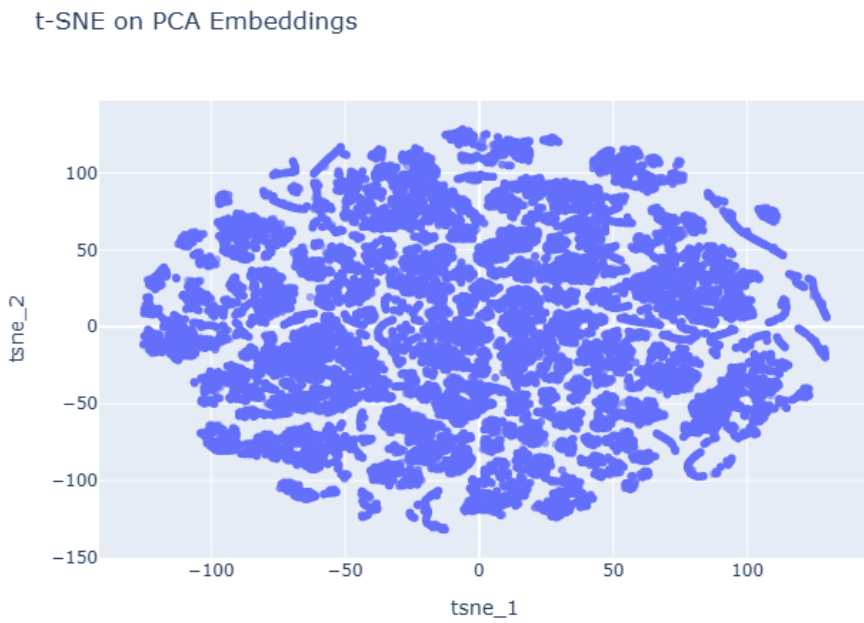

Figure 3: t-SNE scatter on PCA embeddings.

6 Technical Interpretation

1. **Numeric features are compact**: PCA captures 90.12% of numeric-feature variance with 20 components, indicating strong redundancy among rating statistics and genre indicators.
2. **Text features are diffuse**: Truncated SVD reaches 35.72% cumulative variance at 200 components, reflecting higher semantic diversity across tags and titles.
3. **Representation trade-off**: Numeric signals compress well, while text signals preserve more long-tail variation; downstream models should treat these spaces differently.

7 Reproducibility

To rebuild all Week 5 outputs:

```
python3 scripts/build_week05_pipeline.py
```